

ing to hitting a specific room. Soon the next generation technologies will turn the foot soldier, with an advance chip in his boot, into a precision strike weapon, able to navigate without GPS and fire guided bullets at targets like would-be snipers before they have a chance to fire at him.

Types of guided weapons

Different types of guided weapons mainly include anti-radiation weapons, radar-guided weapons, laser-guided weapons, infra-red-guided weapons, wire-guided weapons, beam-riding weapons and GPS/INS-guided weapons.

Anti-radiation weapons. Anti-radiation weapons (ARMs) are designed to target ground based radars. These do so by detecting radio emission from the radars and then homing onto the source of radio emission. These weapons can also be used to target jammers and radios used for communication. Air-to-surface, surface-to-surface, surface-to-air and air-to-air variants of ARMs are in use. A common deployment of these weapons is in specialist aircrafts meant to target ground based radars in suppression of enemy air defence (SEAD) role.

AGM-88 anti-radiation missile (HARM) of the USA is an example of air-to-surface anti-radiation weapon (Fig. 1). Surface-to-surface ARM, such as MM40 Exocet (Fig. 2), employs an active radar seeker, whose receiver component is used to home onto the radar. These missiles are extremely hard to defeat with electronic countermeasures.

Surface-to-air ARMs are used to target airborne early warning (AEW) and airborne warning and control systems (AWACSs). FT-2000 system of People's Republic of China is an example. More recently, air-to-air ARM designs have also begun to appear on the scene, such as the anti-radiation Russian Vympel R-27P. Such missiles do not trigger any radar warning receivers and therefore are relatively immune to countermeasures.



Fig. 3: Hellfire missile

Radar-guided weapons. Radar guidance, which includes semi-active radar guidance and active radar homing, is commonly used in long-range surface-to-air and air-to-air missiles. In the case of semi-active radar guidance, external radar irradiates the target, and the missile-seeker head makes use of the signal reflected off the target to home on to the target.

Active radar-homing missile has a radar transceiver on board the missile, which finds and continuously tracks the target till it hits it. MBDA MICA short- and medium-range surface-to-air and air-to-air missiles, MBDA EXOCET anti-ship missile of France and DRDO-Astra BVRAAM of India are some examples.

Laser-guided weapons. These weapons make use of a laser beam to guide the weapon (bomb, projectile or missile) to precisely hit the target. Most laser-guided munitions, with the exception of laser beam riding, operate on the principle of semi-active laser homing, similar to semi-active radar homing. A laser beam irradiates the intended target—a process called laser designation. It bounces off the target and gets scattered in all directions. The laser seeker in the munition detects the direction of arrival of laser energy and guides it to the target. Targets employ laser-absorbing paints, smoke screens and active protection systems as countermeasures.

Laser-guided munitions are available as canon-launched surface-to-surface projectiles, aerially-delivered bombs, and surface-to-surface, surface-to-air and air-to-surface missiles. Copperhead and Krasnopol

(both canon-launched projectiles), Paveway-II (aerially-delivered bomb) and Hellfire (surface-to-surface and air-to-surface missile) are some examples. Fig. 3 shows Hellfire missile, depicting its inner parts.

Infra-red-guided weapons. Infra-red-guided weapons make use of electromagnetic radiation emitted from the target, predominantly in the infra-red part of the spectrum, to track the target and then home onto it. Such missiles are also referred to as heat-seeking missiles. The seeker head in this case is an infra-red sensor, located on the tip or head of the missile. The infra-red seekers are designed to be sensitive to either 3-5 micron band, in which case these are called single-colour seekers, or 3-5 micron and 8-12 micron bands and referred to as two-colour seekers. Infra-red missiles using two-colour seekers are far more immune to countermeasures like flares.

Another variant of the infra-red-guided missile is the one employing an imaging infra-red (IIR) seeker head, which uses IR/UV focal plane sensor array. Missiles employing IIR seeker heads are far more resistant to countermeasures and are less likely to be fooled into locking onto the Sun's radiation.

IRIS-T manufactured by Diehl BGT Defence as part of Germany-led multinational program is an advanced air-to-air guided missile employing IIR seeker. It has three variants including IRIS-T (air-to-air guided missile), IRIS-TSL (surface-launched medium-range-guided missile) and IRIS-T (surface-launched short-range-guided missile).

Wire-guided weapons. In this case, the missile is guided by electrical signals sent to it through a bundle of wires connected between the missile and the guidance mechanism located near the launch site. The wires reel out behind the weapon as it flies. Wire guidance is commonly used in anti-tank missiles, where its suitability in limited line-of-sight availability is particularly advantageous.